Mixed Numbers & Height Equivalence: 3/4 + 2 1/2

Fraction Addition & Mixed Numbers • Standard: CCSS.MATH.5.NF.A.1

1. MATHEMATICAL CONCEPT & STACK APPROACH: Represent proper and mixed fractions physically with block towers and computationally with two-dot notation.

Fractions can only be added or subtracted when they share a common unit (denominator), and mixed numbers combine whole units with fractional parts. On the stack, two-dot notation ([.] n [.] d) treats each fraction or mixed number as a single unified value on Y and X.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: 84 mm Fraction Towers & Twin Comparator Well

Hands-On Setup: Place three 1/4 blocks (3 × 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm).

Student Prediction (Before Calculator): Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[•]	0.	Leading dot indicates 0 wholes
[3]	0.3	Numerator = 3
[•]	0.3.	Second dot sets fraction entry mode
[4]	3/4	Denominator = 4: displays 3/4 (or 0.75)
[ENTER]	3/4	3/4 lifted to register Y
[2]	2	Whole = 2
[•]	2.	Dot
[1]	2.1	Numerator = 1
[•]	2.1.	Dot
[2]	2 1/2	Denominator = 2: displays 2 1/2
[+]	3 1/4	Plus combines: 3 1/4 (decimal 3.25)!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (The Pharaoh's Bakery & Why the Whole Fraction Tower Stands 84.0 mm Tall)

Four thousand years ago, Egyptian scribes divided 3 loaves of bread fairly among 4 workers by giving each worker 1/2 + 1/4 of a loaf! Because the Whole block in the Fraction Studio is engineered to 84.0 mm (divisible by 1, 2, 3, 4, 6, 7, 8, and 12), a 42 mm (1/2) block plus a 21 mm (1/4) block stacks to 63 mm—matching three 21 mm (3/4) blocks by sight and touch.

Core Investigation (Student Warm-Up A):

Stack a 1/2 tower (42 mm) and a 1/4 tower (21 mm) together. How many 1/4 blocks match their combined height, and what is 1/2 + 1/4?

Fraction Multiplication & Reciprocals

Fraction Multiplication & Inverses • Standard: CCSS.MATH.5.NF.B.4

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Multiplying fractions ((3/4) × (2/3)) scales one rational quantity by another, while dividing by a fraction multiplies by its reciprocal. On the stack, placing the dividend in Y and divisor in X lets you either divide directly or invert X ([1/x]) and multiply.

Blocks & Prediction: Set up 3/4 of a tower (63 mm) and divide into 3 equal parts (21 mm each), taking 2 parts: (3/4) × (2/3). Prediction: 3/4 × 2/3 = 6/12 = 1/2. The resulting block height matches one 1/2 block (42 mm).

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[•] [3] [•] [4] [ENTER] [•] [2] [•] [3] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[•]	0.	Start proper fraction 3/4
[3]	0.3	Numerator 3
[•]	0.3.	Fraction separator
[4]	3/4	3/4 in register X
[ENTER]	3/4	3/4 lifted to register Y
[•]	0.	Start proper fraction 2/3
[2]	0.2	Numerator 2
[•]	0.2.	Fraction separator
[3]	2/3	2/3 in register X
[×]	1/2	Product 6/12 = 1/2 (matches 42 mm block)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Adding numerators and denominators directly (e.g., 1/2 + 1/4 = 2/6) because 6 looks larger.

Hands-On Block Remediation: Place the 1/2 tower (42 mm) + 1/4 tower (21 mm) beside two 1/6 towers (28 mm) in the Fraction Studio comparator lid; the physical height difference proves 1/2 + 1/4 = 3/4 (63 mm).

Misconception 2: Entering a proper fraction like 3/4 as [3] [.] [4] (which is decimal 3.4) instead of using Two-Dot fraction notation [.][3][.][4].

Hands-On Block Remediation: Teach the rule: 'First dot = whole number zero, second dot = fraction bar.' Typing [.][3][.][4] displays exact 3/4 on the stack.

4. EXTENSION MATHEMATICAL INSIGHT

Multiplying fractions ((3/4) × (2/3)) scales one rational quantity by another, while dividing by a fraction multiplies by its reciprocal. On the stack, placing the dividend in Y and divisor in X lets you either divide directly or invert X ([1/x]) and multiply.

Fraction Studio: Height Equivalence & Two-Dot Notation

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: TWO-DOT FRACTION ENTRY & TOWER EQUIVALENCE

- One Whole tower stands 84.0 mm tall, dividing cleanly into halves (42 mm), thirds (28 mm), fourths (21 mm), sixths (14 mm), eighths (10.5 mm), and twelfths (7 mm).
- Twin Comparator Well: Stack equivalent fraction towers side-by-side against the 84.0 mm Whole tower to verify equivalence by sight and touch.
- Two-dot entry format: [WHOLE] [.] [NUMERATOR] [.] [DENOMINATOR] (or [a ^{b}/c] on SC-6).

Key Sequence: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Investigation 1: Add 1/3 + 1/6 by trading 1/3 for two 1/6 blocks, then simplify the sum to lowest terms.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 2: Subtract 1 1/2 - 3/4 by trading the whole and half into fourths (6/4 - 3/4). What fraction remains?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). Which is smaller, and what is 3/8 written as a decimal (3 ÷ 8)?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 4: Find the area of a rectangle measuring 2/3 unit by 3/4 unit by multiplying (2/3) × (3/4) and simplifying.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Unit 06: Fraction Studio: Height Equivalence & Two-Dot Notation — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 4–5

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Stack a 1/2 tower (42 mm) and a 1/4 tower (21 mm) together. How many 1/4 blocks match their combined height, and what is 1/2 + 1/4?

Core Solution: 3/4 (three 21 mm fourth-blocks = 63 mm).

Core Check Keys: [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+]

Extension Challenge (Warm-Up B): Find 2/3 of 3/8 by multiplying (2/3) × (3/8) and simplifying to lowest terms. Which unit fraction tower matches the product?

Extension Solution: 1/4 (6/24 = 1/4, height 21 mm).

Ext. Check Keys: [•] [2] [•] [3] [ENTER] [•] [3] [•] [8] [×]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Add 1/3 + 1/6 by trading 1/3 for two 1/6 blocks, then simplify the sum to lowest terms.

Expected Answer: 1/2 (42 mm)

SC-6 Verification Keys: [•] [1] [•] [3] [ENTER] [•] [1] [•] [6] [+]

Investigation 2: Subtract 1 1/2 - 3/4 by trading the whole and half into fourths (6/4 - 3/4). What fraction remains?

Expected Answer: 3/4 (63 mm)

SC-6 Verification Keys: [1] [•] [1] [•] [2] [ENTER] [•] [3] [•] [4] [-]

Investigation 3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). Which is smaller, and what is 3/8 written as a decimal (3 ÷ 8)?

Expected Answer: 0.375

SC-6 Verification Keys: [•] [3] [•] [8] [FDISP]

Investigation 4: Find the area of a rectangle measuring 2/3 unit by 3/4 unit by multiplying (2/3) × (3/4) and simplifying.

Expected Answer: 1/2 (42 mm)

SC-6 Verification Keys: [•] [2] [•] [3] [ENTER] [•] [3] [•] [4] [×]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.